

RSA Encryption Algorithm

Aim:

To implement RSA asymmetric encryption and decryption algorithm using HTML & JavaScript with the Web Crypto API (RSA-OAEP)

Algorithm:

Key Generation

* Call window.crypto.subtle.generateKey()

specifying:

- RSA-OAEP

- Modulus length : 2048-bit

- Public exponent : 65537

- Hash : SHA-256

Encryption Process:

* Convert plaintext message into a byte buffer using TextEncoder()

* Pass public key and plaintext buffer to window.crypto.subtle.encrypt()

* Encode output ArrayBuffer into Base64 format.

Decryption Process:

* Pass private key and encrypted buffer to window.crypto.subtle.decrypt()

* Decode decrypted ArrayBuffer using TextDecoder()

Output:

1. Key Generation:

Status: RSA Key Pair successfully generated
2048-bit RSA-OAEP.

2. Plaintext input:

Message: "Hallo, RSA Security"

3. Encryption Output (Base64 Ciphertext):

e8a1B9c02F [256-bit] a9F26801

4. Decryption Output:

Decrypted text: "Hallo, RSA Security!"